

practical 78%
oral 83%

ABERDARE TECHNICAL COLLEGE INFORMATION
COMMUNICATION TECHNOLOGY MODULE 2
UNIT: COMPUTER REPAIR AND MAINTENANCE | UNIT
CODE: 0714 351 04A
PRACTICAL 1, TERM 2 | LEVEL 5
Total: 50 marks Oral: 30 marks

ASSESSOR'S OBSERVATION CHECKLIST

Trainer Details

Trainer name	James	Staff/ID no.	20494727
Department	computer sciences	Date of assessment	09/10/2026
Trainer signature	<i>[Signature]</i>	Venue/Workstation	Computer Lab 210

Trainee Details

Trainee name	Collins Ngatho	Admission no.	1087
Class/Course	ICT	Date	9/10/2026
Trainee signature	<i>[Signature]</i>	Workstation no.	82

Instructions to the assessor:

- ✓ Observe the candidate throughout the practical and tick each criterion as it is met.
- ✓ Award marks up to the maximum shown.
- ✓ Part C (oral) is conducted after the practical.
- ✓ Enter marks in the summary table.

Cutting List: Tools and Materials

Items marked Centre are provided by the college. Items marked Student must be brought by the trainee.

No	Item	Quantity	Provided by
Hardware and consumables			
No	Item	Quantity	Provided by
1	Faulty desktop computer with keyboard, mouse, monitor and power cable	1 set	☒ Centre
2	Spare parts tray (1 HDD with OS installed, SATA data and power cables, other spare parts)	1 tray	☒ Centre
3	Diagnostic tools (bootable USB or DVD)	1	☒ Centre

4	Thermal paste	1 tube/sachet	☒ Centre
5	Cable ties and Velcro straps	10 ties, 2 straps	☒ Centre

Cleaning materials

6	Air blower (hand blower or compressed air can)	1	☒ Centre
7	Microfibre cloth	1	☒ Centre
8	Soft anti-static brush	1	☒ Centre
9	Isopropyl alcohol (90% or higher)	1 small bottle	☒ Centre
10	Lint-free swabs/cotton buds	1 pack	☒ Centre
11	Dust mask	1	☒ Student
12	Safety goggles	1 pair	☒ Student

Tools

13	Straight-head (flat) screwdrivers, large and small	1 each	☒ Centre
14	Phillips-head screwdrivers, large and small	1 each	☒ Centre
15	Torx screwdriver (if needed)	1	☒ Centre
16	Anti-static wrist strap	1	☒ Centre

No	Item	Quantity	Provided by
17	Anti-static mat	1	☐ Centre
18	Anti-static bags	2	☒ Centre
19	Needle-nosed pliers	1	☒ Centre
20	Tweezers or part retriever	1	☐ Centre
21	Wire cutters	1	☒ Centre
22	Chip extractor	1	☒ Centre
23	Plastic spudger/opening tool	1	☐ Centre
24	Magnetic parts tray or small labelled containers	1	☐ Centre
25	Flashlight or work lamp	1	☐ Centre

26	Marker pen and masking tape	1 each	☒ Centre
27	E-waste collection box	1 per room	☒ Centre
Documentation and stationery			
28	Computer/motherboard service manual	1	☒ Centre
29	Job card/repair log	1	☒ Centre
30	Pen	1	☒ Student
31	Ruler	1	☒ Student
32	Notepad	1	☒ Student

Part A: Practical Observation Checklist (50 marks)

No	Observation criteria (tick when observed)	Max	Marks	Remarks
Task 1: Preparation and Initial Inspection (6 marks)				
1	☐ Wears anti-static wrist strap clipped to bare metal of the case	2	02	wore anti-static wrist strap - clipped
2	☐ Wears dust mask and safety goggles before blowing dust	1	00	- no dust mask worn
3	☐ Prepares a clean, well-lit workstation and unplugs the power cable before opening the case	1	01	- opened the curtains to ensure the area is well lit
4	☐ Powers on, observes and records symptoms (fan noise, heat, BSOD, boot error, beep codes)	2	02	power on the machine and records all the symptoms

Task 2: Disassembly and Diagnosis (12 marks)

5	☐ Disconnects all external cables and peripherals	1	01	disconnects all external peripherals
6	☐ Removes side panel and photographs or sketches internal layout and cables	2	02	side panel removed and photographed (the system unit)
7	☐ Identifies the faulty component using BIOS/UEFI, cable check and diagnostic tools, and records reasoning	4	03	faulty components - HDD - fans etc (identified and replaced)
8	☐ Disconnects and removes the faulty part correctly, places it in an anti-static bag and labels it	3	02	- faulty part removed - not placed in anti-static bag
9	☐ Keeps screws organised and does not force parts	2	01	- screws not organized

Task 3: Cleaning and Dust Removal (13 marks)

10	☐ Uses air blower correctly and holds fan blades still while blowing	3	03	- used correctly - holds blades while blowing
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No	Observation criteria (tick when observed)	Max	Marks	Remarks
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11	☐ Cleans CPU fan and heatsink fins thoroughly	2	0	Cleaned but not thoroughly
12	☐ Cleans case fans, vents, RAM and expansion slots	2	0	Cleaned fans and vents only
13	☐ Cleans PSU vents from outside only and does not open the PSU	1	0	Cleaned
14	☐ Checks thermal paste, cleans old paste and applies fresh pea-sized amount	3	0.5	check thermal paste
15	☐ Cleans monitor screen and casing with microfibre cloth, with no liquid sprayed on the screen	2	0	Cleaned monitor

Task 4: Installing the Replacement Part (8 marks)

16	☐ Selects the correct replacement and checks compatibility (interface, size, connectors)	2	0	Had (old) selected
17	☐ Mounts the replacement part in its position and secures it with screws	2	0	old mounted
18	☐ Connects data and power cables correctly and firmly	2	0	
19	☐ Ties cables neatly so no fan is blocked	2	0	

Task 5: Reassembly and Testing (8 marks)

20	☐ Refits CPU cooler and fans, connects fan power, and closes the case	2	0	Refits CPU cooler
21	☐ Reconnects peripherals, powers on and confirms POST with no error beeps	2	0	Peripherals reconnected
22	☐ Enters BIOS/UEFI, confirms components are detected and sets the correct boot device	2	0	BIOS/UEFI not accessed
No	Observation criteria (tick when observed)	Max	Marks	Remarks

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23	☐ Verifies the repair (boots, no BSOD in about 10 minutes, fans and temperature normal)	2	2	Repair Verified
Task 6: Disposal and Final Clean-up (3 marks)				
24	☐ States correct disposal of the faulty part (data sanitization, certified e-waste recycler) and places it in the e-waste box	2	2	
25	☐ Returns all tools and leaves the workstation clean	1	1	
TOTAL PRACTICAL		50		

Part B: Product Checklist (10 marks)

No	Product criteria (tick when met)	Max	Marks	Remarks
1	☐ Computer boots to the desktop with no "No bootable device" message	3	3	Boot came up no errors!
2	☐ No BSOD during the test run (about 10 minutes)	2	2	no BSOD after repairs
3	☐ Fans spin freely, are dust-free, and temperature is in the normal range	2	2	fans spin - no noise
4	☐ Monitor screen and casing are clean and streak-free	1	1	cleaned
5	☐ Case closed neatly, no loose or missing screws and cables tidy	1	1	Case/cables closed
6	☐ Job card fully and correctly completed (including the faulty component identified)	1	1	filled
TOTAL PRODUCT		10	10	

Part C: Oral Questions (30 marks)

Ask each question and award up to 2 marks (1 mark per correct key point). Questions 1 to 8 come from Elements 2.1 and 2.2; Questions 9 to 15 are derived from the practical.

No	Question	Expected answer	Max	Marks	Remarks
2.1 Selection of computer components					

No	Question	Expected answer	Max	Marks	Remarks
1	2.1.1 Computer hardware components Name four internal hardware components of a desktop computer and state the function of each.	Any four with function (1 mark per two correct pairs): CPU (processes instructions); RAM (temporary working memory); motherboard (connects all components); HDD/SSD (permanent storage); PSU (supplies power); graphics card (video output); cooling fan/heatsink (removes heat).	2	2	skipped 4 components (internal) and state their functions
2	2.1.1.1 Factors to consider State four factors you consider when selecting a replacement component such as a hard disk.	Any four (half mark each): compatibility (interface e.g. SATA, form factor); capacity; speed/performance; power requirements; brand and quality; warranty; cost; availability.	2	2	skipped the factors/standards to consider
3	2.1.1.2 Parts acquisition State two ways of acquiring replacement computer parts and one precaution to take.	Ways: authorised dealers/retailers, online suppliers, manufacturer/vendor, salvaging from old machines, client supplying the part. Precautions: confirm genuine/new part, check warranty, keep receipt, test before fitting.	2	2	skipped 4 way.
2.2 Tools for repair and replacement					

No	Question	Expected answer	Max	Marks	Remarks
4	2.2.1.1 Straight-head screwdriver State one use of a large and a small straight-head (flat) screwdriver.	Large: loosening/tightening bigger slotted screws, e.g. case screws, prying covers. Small: small slotted screws in laptops/devices, releasing connector tabs and jumpers.	2	1	stated but no explanation
5	2.2.1.2 Phillips-head screwdriver When would you use a large and a small Phillips-head screwdriver?	Large: case panels, PSU, drive-bay and motherboard standoff screws. Small: small drive, RAM and laptop screws, expansion cards. Matching size prevents stripped screw heads.	2	1	
6	2.2.1.3 Tweezers/part retriever What is a tweezers or part retriever used for in computer repair?	Picking up and placing small parts (screws, jumpers) and retrieving items dropped in tight spaces without touching the board.	2	1	incomplete explanation
7	2.2.1.4 Needle-nosed pliers State two uses of needle-nosed pliers.	Any two: holding or bending small components, straightening bent pins, pulling out jumpers, reaching tight spaces, gripping small nuts/standoffs.	2	1	
8	2.2.1.5 Wire cutters and 2.2.1.6 Chip extractor State the use of wire cutters and of a chip extractor.	Wire cutters: cut cable ties and trim wires/cables. Chip extractor: safely removes IC chips from sockets without bending the pins (1 mark each).	2	1	stated the use of wire cutters only
Questions derived from the practical					

No	Question	Expected answer	Max	Marks	Remarks
9	Why did you wear an anti-static wrist strap and unplug the computer before opening it?	Strap discharges static (ESD) that can damage chips; unplugging prevents electric shock and short-circuit damage.	2	2	Reasons stated accurately.
10	Explain how you identified the faulty component in this computer.	Linked symptoms (BSOD, "No bootable device") to storage; checked drive detection in BIOS/UEFI; checked cable connections; ran diagnostic/disk health tool; ruled out other parts.	2	2	Explained correctly.
11	Why did you hold the fan blades still when using the air blower?	Blowing makes the fan over-spin and can damage the bearings and generate voltage that harms the board.	2	2	Justification given.
12	Why did you apply thermal paste when refitting the CPU cooler?	Fills microscopic gaps for efficient heat transfer from CPU to heatsink; prevents overheating.	2	2	Explained.
13	How does dust accumulation cause overheating and how can it be prevented?	Dust blocks airflow and insulates heatsink so heat builds up. Prevention: regular cleaning, dust filters, clean room, proper case ventilation.	2	2	
14	What causes the BSOD and "No bootable device" error, and how did your repair fix it?	Failing or disconnected drive, corrupted system files or overheating. Replacing the drive with a working one containing a good operating system fixed the problem.	2	2	stated accurately.

No	Question	Expected answer	Max	Marks	Remarks
15	How did you dispose of the faulty part, and why is it not put in the ordinary bin?	Data wiped/destroyed, handed to certified e-waste recycler/collection box. Contains toxic metals (lead, mercury, cadmium) that pollute soil and water.	2	2	
TOTAL ORAL			30		

25

Summary of Marks

Section	Max marks	Marks awarded	Percentage
A. Practical observation	50	39	78%
B. Product	10	10	
C. Oral	30	25	83%
GRAND TOTAL	90		

Competence decision:

☒ Competent

☐ Not yet competent

Assessor's comments:

Performance Good

Candidate signature / Date	Assessor signature / Date
 9/10/2026	 10/09/2026